

Apostol Chapter 1 - Exercise 1

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1. If $(a, b) = 1$ and if $c|a$ and $d|b$, then $(c, d) = 1$

Obviously, $1|c$ and $1|d$, i.e, 1 is a common divisor of c and d .

Let $e \in \mathbb{Z}$ be such that $e|c$ and $e|d$.

Since $e|c$ and $c|a$, by transitivity, we have that $e|a$. Similarly, from $e|d$ and $d|b$, we get that $e|b$. Then, by definition of (a, b) , we conclude that $e|(a, b)$, but $(a, b) = 1$, then $e|1$.

Therefore, $(c, d) = 1$.